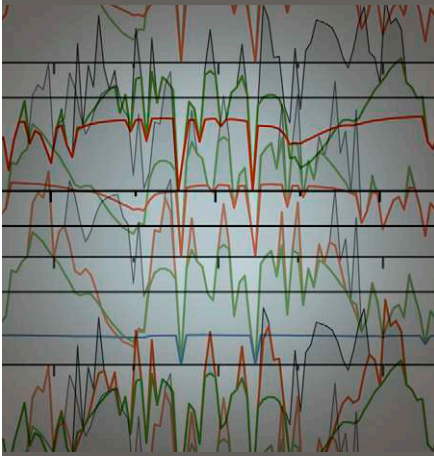


Original Research

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A dual-continuum model of soil-moisture dynamics that incorporates water uptake by plants via a mechanism driven by a difference in water potential is presented. The model is used to predict the root-mediated redistribution of water in a forest soil and compared with extensive field measurements.

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Macroscopic Modeling of Plant Water Uptake in a Forest Stand Involving Root-Mediated Soil Water Redistribution

One of the principal components of mass exchange within the soil–plant–atmosphere system is soil water extraction by plant roots. Adequate evaluation of water extraction is a prerequisite for correct predictions of plant transpiration and soil water distribution in the root zone. The main objective of the present study is to contribute to the development of sufficiently realistic, yet algorithmically simple models of water exchange between soil and plant roots applicable for numerical simulation of soil water responses to atmospheric forcing. In our case, a simple macroscopic, vertically distributed plant root water uptake approximation based on a traditional water-potential-gradient (WPG) formulation was adopted and implemented in a one-dimensional dual-continuum model of soil water flow based on the Richards' equation. This combined model was used to simulate soil water movement at a forested site. The results were compared with observations (sap flow, soil water pressure, and soil water content) as well as with simulations produced using the standard semi-empirical model of Feddes. Principal aspects of the WPG prediction, such as root-mediated soil water redistribution, compensation for local water scarcity, and transpiration reduction, are exposed and discussed.

Abbreviations: HFD, heat field deformation; RWU, root water uptake; SPA, soil–plant–atmosphere; WPG, water-potential-gradient.

The correct prediction of plant root water uptake (RWU) is increasingly important in hydrological, climatological, agricultural, and ecological studies dealing with vegetation related processes at the land–atmosphere interface. In such studies, some form of RWU model is needed to obtain a quantitative link between soil water and atmospheric water in the soil–plant–atmosphere (SPA) continuum. In numerical models of soil water flow, the plant root water uptake model is usually implemented in governing equations of soil water flow via a macroscopic sink term (e.g., Hillel et al., 1976; Feddes et al., 1978).

One important aspect of the spatial distribution of RWU is the so-called compensation mechanism, which allows plants to compensate for dry soil layers by increasing uptake from wet layers. The compensation mechanism was documented in a number of experimental studies (e.g., Nnyamah and Black, 1977; Hasegawa and Yoshida, 1982) and incorporated into several numerical models, such as CERES (Ritchie, 1985), MACRO (Jarvis, 1994), HYDRUS-1D (Simunek et al., 2005), and SWAP (Kroes et al., 2008).

Nadezhdina et al. (2010) demonstrated on an extensive set of experimental data that roots may significantly affect the distribution of soil moisture by providing a hydraulically active connection between different soil horizons. This effect—sometimes termed hydraulic lift—is not accounted for by the most commonly used RWU sink term algorithms. Hydraulic lift (and/or descent) may be particularly important in dry periods with limited soil water availability. The first evidence of such root behavior was reported by Richards and Caldwell (1987). However, a thorough long-term assessment of the phenomenon under different vegetation conditions has not yet been provided.

Hillel et al. (1976) presented a macroscopic model of soil moisture extraction by roots based on an Ohm's law analogy (first used in the RWU context by van den Honert, 1948) and tested its validity by conducting potted ryegrass transpiration experiments under controlled environmental conditions (Belmans et al., 1979; Feyen et al., 1980). Their model included an implicit compensation mechanism and potentially allowed negative RWU

rates; however, the latter option was not utilized. Belmans et al. (1979) and Feyen et al. (1980) pointed out the importance of radial root hydraulic resistance in RWU modeling, as did Molz (1981) in his review of different RWU concepts. Guswa (2012) used a similar RWU model (based on water potential gradient and combined root and soil resistances) to investigate the effects of canopy interception and root uptake on soil water balance. Jarvis (1989) and Li et al. (2001, 2006) presented a RWU model with an empirical compensation mechanism.

de Jong van Lier et al. (2008) suggested a physically-based RWU model with an implicit compensation mechanism, such that reductions in RWU from dry soil layers can be compensated by an increased RWU from wetter layers. Their model could potentially be used to simulate negative RWU rates, allowing for root mediated redistribution of soil water. However, the authors decided [as did Belmans et al. (1979)] to prevent the negative RWU rates by forcing the respective root extraction intensities to zero. Unlike in Hillel et al. (1976), the radial root resistance in the de Jong van Lier et al. (2008) model was assumed negligible, which seems to be a highly unrealistic assumption. This limitation was recognized by Durigon et al. (2012) in their follow-up study.

Jarvis (2010, 2011) compared the macroscopic plant water uptake models of Jarvis (1989) and de Jong van Lier et al. (2008). He pointed out a conceptual similarity between the two models, and provided illustrative simulations of the compensatory RWU effects in two case studies considering different scales and types of vegetation. In concluding, he suggested that more research is needed to test alternative RWU concepts against experimental data. Following the concept of Jarvis (1989), Simunek and Hopmans (2009) proposed an RWU model involving compensation for active nutrient uptake, implemented similarly to water uptake. The proposed root water and nutrient uptake model was illustrated by several hypothetical example problems. de Willigen et al. (2012) compared the performance of four different RWU models of varying complexity in numerical experiments under extreme atmospheric and soil water conditions.

In this article, we apply a simple physically-based macroscopic RWU model [pointing back to the Hillel et al. (1976) concept] to simulate soil water responses to transpiration in a forest site located in a temperate humid climate of central Europe. The suggested modeling approach is implemented in a one-dimensional dual-continuum model of soil water flow based on the Richards' equation. The model was used to simulate soil water movement in situ over the period of one vegetation season. The main objective of this study is to test the ability of the presented RWU model to simulate the observed soil–plant–atmosphere interactions, and to compare the differences between empirical and more physically-based root water uptake modeling approaches (accommodated in the same soil water flow model). The model is applied in a case study where observations of the relevant micrometeorological and

hydrological variables were available in a high resolution over a large part of a vegetation season.

Material and Methods

Experimental Site

The Liz experimental site is situated in the headwater area of the Volynka river basin in Bohemian Forest, south-west Bohemia (total area 0.99 km², average altitude 941 m above sea level, average annual precipitation 861 mm, average annual temperature 6.5°C). The prevailing soil type in the headwater area is oligotrophic forest Eutric Cambisol developed on biotite paragneiss bedrock. The area is mostly covered by spruce and beech forest. The site is equipped with two automatic monitoring stations, located in a small meadow and at the nearby spruce forest floor. The stations are used for continuous measurements of air and soil temperatures, soil water pressure (tensiometer sensors T8 and T4e, UMS GmbH, München, Germany), soil water content (FDR probes model CS616, Campbell Scientific Inc., UK), and precipitation (throughfall and stemflow in the forest location). Complete micrometeorological data are available for the meadow location. The micrometeorological conditions are monitored continuously with a 15-min time step. Variables utilized in the present study include rainfall intensity, air temperature, air humidity, barometric pressure, wind speed, and radiation. Sap flow in spruce trees [*Picea abies* (L.) Karst.] is measured by the heat field deformation method (HFD, Nadezhdina et al., 1998). The majority of trees at the spruce forest location are 90 yr old with a height of about 28 m and a diameter of about 40 cm (Fig. 1). Multi-point HFD sensors were used for measuring sap flow radial profiles in six tree stems and single-point sensors were used for selected principal root branches. The HFD installation at one of the trees was complemented with a set of soil water content and soil water pressure sensors inserted in three depths below the soil surface within a radial distance of 1.5 m from the tree stem.



Fig. 1. The Liz experimental site (spruce forest location).

The forest is well supplied with water throughout the whole vegetation season, except for relatively short irregularly occurring dry spells in summer. According to Prazak et al. (1996), limited moisture conditions at Liz lead to reduced actual transpiration rates (compared to the potential rates associated with the atmospheric demand) when the average soil water pressure head in the root zone drops below about -600 cm. The comparison of the long-term catchment water balance components and the calculated cumulative potential transpiration rates (based on micrometeorological observations) indicate moderate reduction of transpiration—less than 20% over the vegetation season in most years.

Soil Water Flow Model

Transient vertical flow of water in a non-homogeneous soil profile with preferential pathways can be described by a one-dimensional dual-continuum model. The approach followed in our study is based on the dual-porosity concept of Gerke and van Genuchten (1993). The dual-continuum model utilizes a set of two Richards' equations, which can be written in the following form (Vogel et al., 2010b):

$$\frac{\partial w_i \theta_i}{\partial t} = \frac{\partial}{\partial z} \left(w_i K_i \left(\frac{\partial h_i}{\partial z} + 1 \right) \right) - w_i S_i - \Gamma_i \quad [1]$$

with coupling occurring through the soil water transfer term Γ_i (s^{-1}):

$$\Gamma_f = -\Gamma_m = \alpha_{ws} K_{ar} (h_f - h_m) \quad [2]$$

In these equations, $i = m$ for the soil matrix domain, $i = f$ for the preferential flow domain, h_i is the domain specific soil water pressure head (m), θ_i is the volumetric soil water content ($m^3 m^{-3}$), and K_i is the unsaturated hydraulic conductivity ($m s^{-1}$). The terms w_m and w_f are volumetric fractions of the respective flow domains (soil matrix or preferential), S_i is a sink term (s^{-1}) used to account for root water uptake, α_{ws} is the interdomain soil water transfer coefficient at saturation ($m^{-1} s^{-1}$), K_{ar} is the relative unsaturated hydraulic conductivity of the interface between the flow domains, t is time, and z is the vertical coordinate (positive upward). The soil water transfer term, Γ_i , is defined as the volume of water moving from the preferential flow domain to the soil matrix domain per unit bulk volume of the medium per unit time. It is dynamically varying proportionally to the water pressure difference between the soil matrix and the preferential flow domain. The dual set of Richards' equations presented is solved numerically by the computer code S1D (Vogel et al., 2010a, 2010b).

When root water uptake is to be accounted for, spatiotemporal distribution of the actual root water uptake rate to be applied via the sink term, S , has to be evaluated. This is done based on prescribed values of the potential transpiration rate, T_p , using a selected root water uptake model.

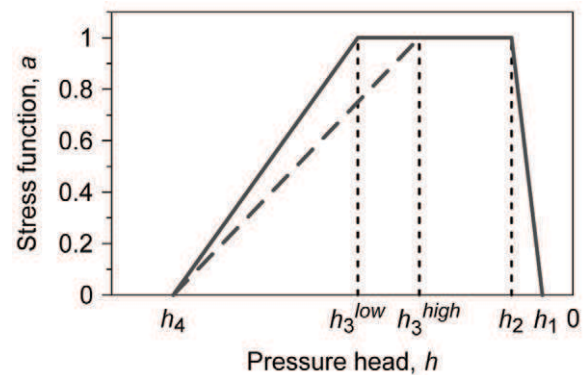


Fig. 2. Plant water stress function of Feddes et al. (1978). The position of h_3 depends on the intensity of potential transpiration and is interpolated in the interval (h_3^{low}, h_3^{high}) , where the superscripts “low” and “high” refer to the magnitude of potential transpiration (often taken as $T_p^{low} = 0.1$ and $T_p^{high} = 0.5$ cm d $^{-1}$).

Feddes' RWU Model

Feddes et al. (1978) proposed the following approximation of RWU rate S as a function of the soil water potential:

$$S(h) = a(h) S_p \quad [3]$$

where $a(h)$ is the prescribed dimensionless function of pressure head h ($0 \leq a(h) \leq 1$), called plant-water-stress function, and S_p is the potential water uptake rate in the root zone (s^{-1}) related to the potential transpiration demand of vegetation under actual microclimatic conditions.

The water stress function recommended by Feddes et al. (1978) is shown in Fig. 2. The pressure head interval (h_3, h_2) , within which $a(h) = 1$, defines the range of optimal soil water conditions for plant transpiration (no water stress). For soil moisture conditions outside this interval, RWU reduction is prescribed. The interval (h_2, h_1) defines the range of reduced transpiration due to the lack of aeration in the root zone and h_4 is the plant wilting point. The potential water uptake rate, S_p , generally varies in space. For modeling purposes, this variation is usually related to the active root distribution. In the case of the potential RWU rate being equally distributed over the root zone, S_p becomes according to Feddes et al. (1978):

$$S_p = \frac{I}{z_R} T_p \quad [4]$$

where T_p is the potential transpiration rate ($m s^{-1}$), and z_R is the depth of the root zone (m).

To account for a non-uniform distribution of the potential water uptake rate over the root zone, Eq. [4] has been generalized by e.g., Vogel (1987):

$$S_p = b(z)T_p \quad [5]$$

where $b(z)$ is the normalized potential RWU distribution function (m^{-1}) that fulfills the normalization condition:

$$b(z) = \frac{b'(z)}{\int_{\Omega_R} b'(z) dz} \quad [6]$$

where Ω_R is the region occupied by the root zone, and $b'(z)$ is a prescribed distribution function, often associated with the root length density. Normalizing the potential RWU distribution according to Eq. [6] ensures that $b(z)$ integrates to unity over the root zone.

The actual water uptake distribution is obtained by substituting Eq. [5] into Eq. [3]:

$$S(z) = a(b(z))b(z)T_p \quad [7]$$

whereas the actual transpiration rate, T_a , is obtained by integrating Eq. [7] as follows:

$$T_a = T_p \int_{\Omega_R} a(b) b(z) dz \quad [8]$$

It is important to note that the Feddes' RWU model, as well as other RWU models discussed in this paper, assume quasi-steady state continuity equation for plant xylem water, i.e.:

$$T_a = \int_{\Omega_R} S dz \quad [9]$$

which is thus a principal simplifying assumption of these models.

A disadvantage of the Feddes' RWU model is that reduced uptake from dry soil layers cannot be compensated by increasing uptake from wetter layers. There are also no provisions allowing simulation of the soil water redistribution through the root xylem (involving negative RWU rates). These shortcomings are being addressed in the RWU model presented below.

RWU Model Based on Water Potential Gradient

At the microscopic scale, the extraction of soil water from the root surface has been traditionally described by analogy to Ohm's law of electric current (e.g., van den Honert, 1948; Kramer, 1983; Kramer and Boyer, 1995); that is,

$$q = \frac{H_{\text{surf}} - H_{\text{root}}}{r_{\text{root}}} \quad [10]$$

where q is the local extraction rate ($m s^{-1}$), H_{root} is the water potential in the root xylem (m), H_{surf} is the water potential at the soil-root interface (m), and r_{root} is the radial hydraulic resistance of the root tissues (s).

The microscopic RWU can be upscaled to the macroscopic level using the specific active root surface σ (m^{-1}):

$$S = \frac{\sigma}{r_{\text{root}}} (H_{\text{surf}} - H_{\text{root}}) \quad [11]$$

where all variables are interpreted as macroscopic quantities, i.e., averaged within a representative elementary volume, and S is the macroscopic rate of extraction of water from a unit volume of soil (s^{-1}).

Macroscopic flow of water from the bulk soil to the active root surface can be described in a similar way, i.e.:

$$S = \frac{\sigma}{r_{\text{soil}}} (H_{\text{soil}} - H_{\text{surf}}) \quad [12]$$

where H_{soil} is interpreted as the water potential of bulk soil and r_{soil} is soil resistance to water extraction by roots. This equation can also be viewed as an application of Darcy's law.

Combining Eq. [11] with Eq. [12] gives:

$$S = \frac{\sigma}{r_{\text{soil}} + r_{\text{root}}} (H_{\text{soil}} - H_{\text{root}}) \quad [13]$$

According to this equation, the macroscopic root water uptake is directly proportional to the gradient of water potential between the bulk soil and the root xylem and indirectly proportional to the combined hydraulic resistances of soil and root tissues. All variables in Eq. [13] can vary throughout the root zone, albeit in different degrees.

Since the longitudinal resistance in the root xylem is expected to be significantly lower than the radial resistance, H_{root} can be assumed approximately constant throughout the root system. Neglecting also the r_{root} variations along roots then leads to:

$$S(z) = \frac{\sigma(z)}{r_{\text{soil}}(z) + r_{\text{root}}} [H_{\text{soil}}(z) - H_{\text{root}}] \quad [14]$$

which is the basis of the RWU model tested in this paper. Equation [14] is an alternative to Eq. [7], the Feddes' RWU model.

For this model to be used, the soil-root system characteristics σ , r_{soil} , and r_{root} need to be quantified. We assume that the effective soil resistance can be expressed as:

$$r_{\text{soil}}(z) = \frac{\lambda(z)}{K(z)} \quad [15]$$

where $\lambda(z)$ is the characteristic length (m) associated with the transport of water from the bulk soil to the root surface, i.e., the distance from the root surface at which the water content is equal to the bulk soil water content. The length $\lambda(z)$ is approximated as (de Jong van Lier et al., 2006):

$$\lambda(z) \approx 0.53 r_m(z) \quad [16]$$

where $r_m(z)$ is the rhizosphere radius (m), which can be determined from the root length density (e.g., de Willigen and van Noordwijk, 1987):

$$r_m(z) = \frac{1}{\sqrt{\pi R(z)}} \quad [17]$$

The specific root surface is estimated from:

$$\sigma(z) = 2\pi r_0 R(z) \quad [18]$$

where $R(z)$ is the root length density (m^{-2}) and r_0 is the average active root radius (m).

As stated in Eq. [14], the actual value of RWU rate S depends on the soil water status, expressed in terms of soil water potential H_{soil} , and the value of root water potential H_{root} , which varies in response to changes of transpiration. Therefore, the evaluation of S has to be preceded by the determination of H_{root} . In our present model, similarly to the algorithm suggested by Hillel et al. (1976), H_{root} is first adjusted so as to maintain the transpiration at its potential rate T_p . The adjustment is no longer valid when the

required H_{root} drops below a threshold value H_{lim} . In that case, H_{root} is kept equal to H_{lim} and the actual transpiration becomes reduced, i.e., $T_a < T_p$. More specifically, H_{root} is calculated from Eq. [9] and Eq. [14], assuming $T_a = T_p$:

$$H_{\text{root}} = \frac{\int_{\Omega_R} \frac{\sigma}{r_{\text{soil}} + r_{\text{root}}} H_{\text{soil}} dz - T_p}{\int_{\Omega_R} \frac{\sigma}{r_{\text{soil}} + r_{\text{root}}} dz} \quad [19]$$

If H_{root} becomes lower than H_{lim} , then the value resulting from Eq. [19] is rejected and replaced by H_{lim} . The actual (reduced) transpiration is then calculated as:

$$T_a = \int_{\Omega_R} \frac{\sigma}{r_{\text{soil}} + r_{\text{root}}} (H_{\text{soil}} - H_{\text{lim}}) dz \quad [20]$$

The above RWU model parameters are fully determined when the following soil and root system properties are quantified: (i) root length density $R(z)$, (ii) average active root radius r_0 , (iii) radial root resistance r_{root} , (iv) threshold value of root water potential H_{lim} , and (v) unsaturated hydraulic conductivity of soil $K(h)$.

Model Application

The soil water flow and RWU model was used to simulate soil water dynamics at the Liz forest site during the period of April 2010 through September 2010. The soil profile was schematized as a vertical one-dimensional 70-cm-deep soil column. For the purpose of numerical modeling, the soil profile was divided into four layers distinguished by different soil hydraulic properties—the organic litter layer and three mineral soil layers (Table 1). The soil hydraulic parameters were evaluated using combined information on retention data (Snehota et al., 2009), measured by standard laboratory methods (sand tank and a pressure plate apparatus), and simultaneous long-term monitoring of soil water pressures and water contents in situ. Saturated hydraulic conductivities

Table 1. Soil hydraulic parameters.

Domain	Layer	Depth	$\theta_r^\dagger$	θ_s	α_{VG}	n_{VG}	K_s	b_s
		cm			cm^{-1}		cm d^{-1}	cm
SM	1	0–8	0.001	0.630	0.030	2.00	1000	0.00
	2	8–20	0.001	0.503	0.042	1.45	150	–2.00
	3	20–55	0.059	0.415	0.038	1.50	20	–2.25
	4	55–70	0.154	0.422	0.034	1.79	1	–2.50
PF		0–70	0.010	0.600	0.020	3.00	5000	0.00

$^\dagger \theta_r$ and θ_s are the residual and saturated soil water contents, K_s is the saturated hydraulic conductivity, b_s is the air-entry value, and α_{VG} and n_{VG} are empirical parameters, determining the shape of retention function (van Genuchten, 1980).

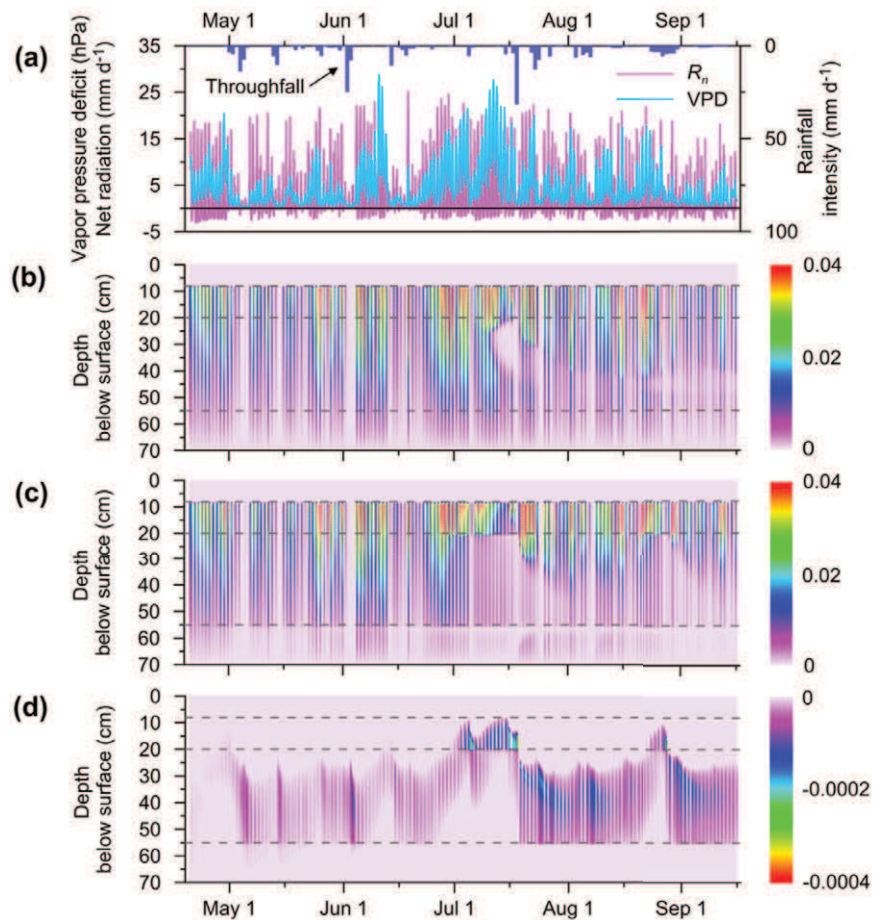


Fig. 3. Observed variations of selected micrometeorological variables together with simulated spatiotemporal distributions of root water uptake rates during vegetation season 2010: (a) hourly net radiation, hourly vapor pressure deficit and daily precipitation, (b) root water uptake (RWU) rates predicted by the Feddes model, (c) positive daylight RWU rates predicted by the water-potential-gradient (WPG) model, and (d) negative RWU rates predicted by the WPG model overnight and during wet canopy events. The net radiation is expressed in units of equivalent evaporation (energy divided by the density of water and the latent heat of vaporization).

were determined by means of ponded infiltration tests and tension disk infiltrometers. The need for a dual-continuum model of soil water flow stemmed from several previous studies conducted at the same watershed under similar soil conditions (e.g., Vogel et al., 2003; Cislerova, 2005). Including the preferential flow domain in the hydraulic model allowed us to match the observed soil water conditions, while keeping the soil matrix hydraulic characteristics as detected with the independent soil tests. The volumetric fraction of the preferential flow domain was estimated as $w_f = 0.05$. The soil water transfer coefficient at full saturation (used by the soil-water-flow model to evaluate the water transfer between flow domains) was assumed to vary linearly between the soil surface and the lower boundary (it was set equal to 0.2 at the soil surface and $0.005 \text{ cm}^{-1} \text{ d}^{-1}$ at the lower boundary of the flow domain). Sensitivity of the dual-continuum model to changes in parameters related to preferential flow was tested by Dohnal et al. (2012).

Precipitation throughfall data, measured directly on-site, were used to define the upper boundary condition (Fig. 3a). Observed

soil water pressure was applied as the bottom boundary condition. Potential transpiration of spruce trees was estimated by means of the Penman–Monteith equation (Monteith, 1981; Votrubova et al., 2012). Transpiration was assumed to drop to zero overnight, as well as during rainfall events (based on precipitation data collected at a nearby meadow site). Dry canopy condition was assumed whenever zero precipitation was recorded. A more sophisticated interception model was not applied because of relatively high uncertainty associated with the parameters controlling the interception storage at the site of interest. We assumed that roots extract water from the soil matrix only (i.e., $S = S_m$, $S_f = 0$).

Two alternative approaches were used to simulate the plant root water uptake: the Feddes' approach and the water-pressure-gradient approach (abbreviated as the WPG model). The Feddes' water stress function parameters were assigned the following values: $h_1 = h_2 = 0$, $h_3^{\text{high}} = -6 \text{ m}$, $h_3^{\text{low}} = -12 \text{ m}$, and $h_4 = -150 \text{ m}$. The values are in good agreement with those reported by Kroes and van Dam (2003) for agricultural crops, as well as those determined by Vogel

et al. (2011) for plant and soil conditions similar to the Liz site. The vertical RWU distribution function, $b(z)$, was assumed identical to $R(z)$ function prescribed for the WPG model (see below).

For the WPG root water uptake model, we estimated a linear decrease of root length density in agreement with the values reported by Borja et al. (2008) for *Picea Abies*, i.e., $R(z) = 0.1 \text{ cm}^{-2}$ near the soil surface (at the depth of 8 cm) and $R(z) = 0$ at the lower boundary of the root zone (at the depth of 70 cm). No active roots were assumed in the uppermost 8-cm layer (organic litter). The average active root radius was set equal to a constant value of $r_0 = 0.005 \text{ cm}$ (as in de Jong van Lier et al., 2008), and the threshold root xylem water potential to a value of $H_{\text{lim}} = -150 \text{ m}$ (in accordance with h_4 in Feddes' RWU model). The radial hydraulic resistance of the root tissues was adjusted to $r_{\text{root}} = 1000 \text{ d}$; the reasoning leading to this value is given below.

Results and Discussion

Simulated spatiotemporal distributions of root water uptake by spruce trees during the 2010 vegetation season are presented in Fig. 3. The precipitation data together with the key micrometeorological variables affecting the magnitude of transpiration (Fig. 3a) indicate that the soil was well supplied with water throughout the season studied, with two drier periods in the first half of July and the second half of August. That is also reflected in the soil water pressure and soil water content data presented in Fig. 4 and 5. The results obtained using either the Feddes or WPG root water uptake models differ particularly in (and soon after) the two dry periods. The differences are due to the contrasting nature of RWU distribution within the root zone (illustrated in Fig. 6)

and different approaches to transpiration reduction. In the WPG model, the shape of the RWU distribution is determined partly by spatial variations of the actual water availability (expressed by the coefficients in Eq. [14]) and partly by the soil water potential profile, since the actual RWU rate is proportional to the water potential differences between the soil and the root xylem (Fig. 7b). The predicted actual transpiration rate is limited by the permissible range of root water potential. In what follows, we first compare simulation results obtained with the two RWU models and then discuss the WPG model predictions in detail.

Compensation for Local Water Scarcity and Transpiration Reduction

As shown in Fig. 6, the Feddes sink term extracts water according to the preset geometric pattern (in our case the triangular root length density function). The local uptake rates are either equal to the potential rates or reduced when the local soil water pressure falls below the preset limits. Thus, local water scarcity leads to a reduction in the transpiration rate. The WPG model distributes the RWU rates according to the actual availability of water in the soil profile, based on local potential gradients and resistances. If soil is well supplied with water (r_{soil} is negligible relative to r_{root}) and the soil water status is close to equilibrium (i.e., the soil water potential approaches the same value in all soil layers), the vertical RWU distribution converges to the shape defined by the root length density function. In the WPG model, the spatial distribution of water extraction always reflects the soil water availability. Water shortage at one region is compensated for by increased extraction at other region(s) where water is more easily available. Reduction in the transpiration rate is predicted only if the transpiration demand cannot be met without the root water potential falling below the limiting value H_{lim} .

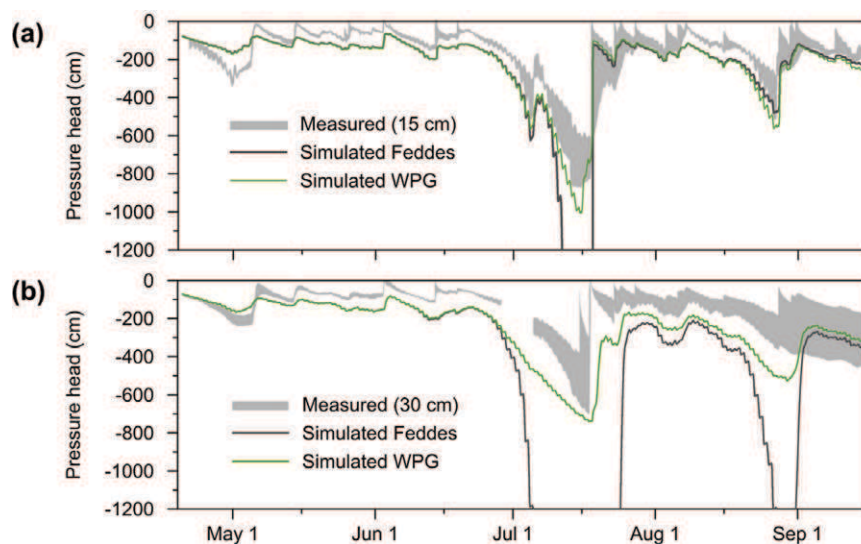


Fig. 4. Pressure head responses at the depth of: (a) 15 cm below the soil surface, and (b) 30 cm below the soil surface. The results of two simulation scenarios are shown. The scenarios are based on the traditional Feddes approach and the water-potential-gradient (WPG) approach, respectively. The soil water pressure heads in the Feddes model decreased down to $-4,700$ and $-14,800 \text{ cm}$ at the depth of 15 cm and 30 cm, respectively. The shaded area represents an envelope of the available tensiometric data.

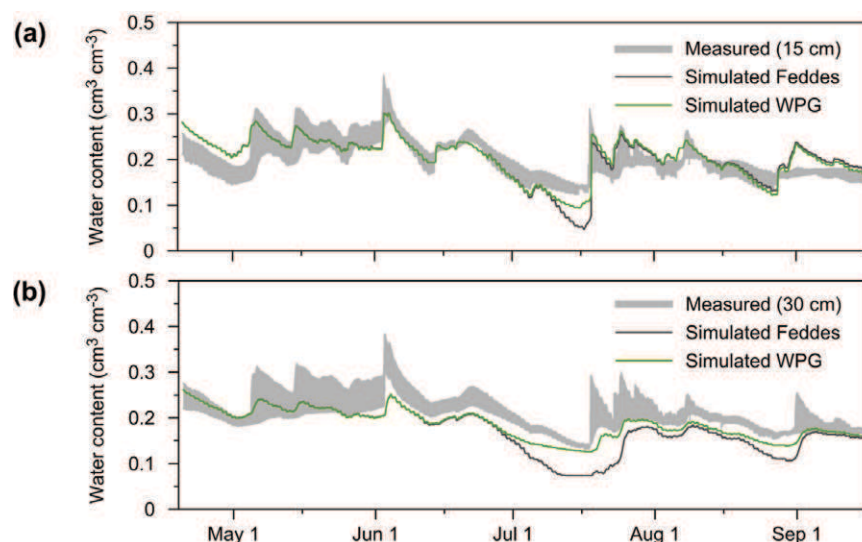


Fig. 5. Soil water content responses at the depth of: (a) 15 cm below the soil surface, and (b) 30 cm below the soil surface. The results of two simulation scenarios are shown. The scenarios are based on the traditional Feddes approach and the water-potential-gradient (WPG) approach, respectively. The shaded area represents an envelope of the available soil water content data.

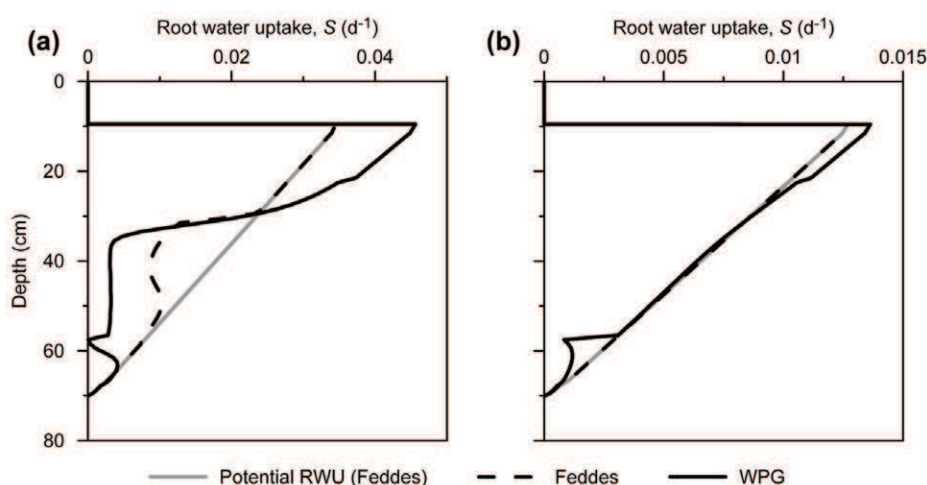


Fig. 6. Comparison of the typical root water uptake profiles predicted by the water-potential-gradient (WPG) model and Feddes' model: (a) profiles corresponding to reduced transpiration under dry conditions (July 20), (b) situation with unlimited transpiration $T_a = T_p$ under wet conditions and near-equilibrium soil water status (May 27). The potential root water uptake (RWU) rate is calculated according to Eq. [5].

As shown in Fig. 3b and 3c, both models predicted reduced extraction from the third soil layer; however, in case of the WPG model, the reduction is partly compensated by increased uptake from the second layer. At the beginning of the simulated period, the RWU rates predicted by both models are very similar; the only noticeable difference relates to the bottom soil layer in which the low soil hydraulic conductivity results in reduced RWU predicted by WPG model. The two models' results start to diverge substantially with the high transpiration demand at the end of June. As soil water content is lower compared to the earlier high-transpiration-demand period at the beginning of June (Fig. 8), the H_{root} in the

WPG model reaches the H_{lim} value and the transpiration rate becomes reduced. At the same time, the vertical distribution of RWU is increasingly influenced by the vertical variation of soil properties; this is demonstrated as an increase of RWU in the second layer and decrease in the third layer (Fig. 3c). The same development is repeated during the second dry spell in the second half of August; in this case, the diversion from the linear RWU distribution is even magnified due to drier conditions at the deeper part of the soil profile (the third layer). Feddes' model starts reducing the transpiration later and the reduction is directly related to the change of the vertical RWU distribution pattern (Fig. 3b). The

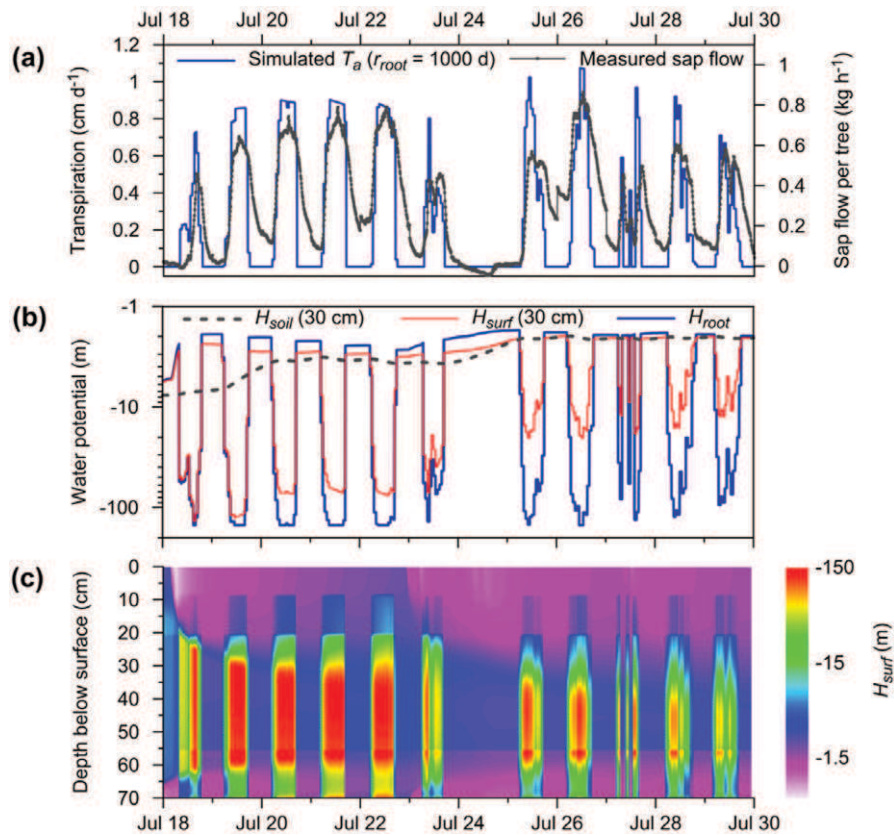


Fig. 7. Simulated diurnal and nocturnal changes of key water-potential-gradient (WPG) root water uptake model variables for the selected 12-d period: (a) actual transpiration – compared with the observed sap flow rates, (b) water potentials in root xylem, at root surface and in bulk soil at the depth of 30 cm below surface, and (c) spatiotemporal distribution of the root surface water potential (calculated from Eq. [11]).

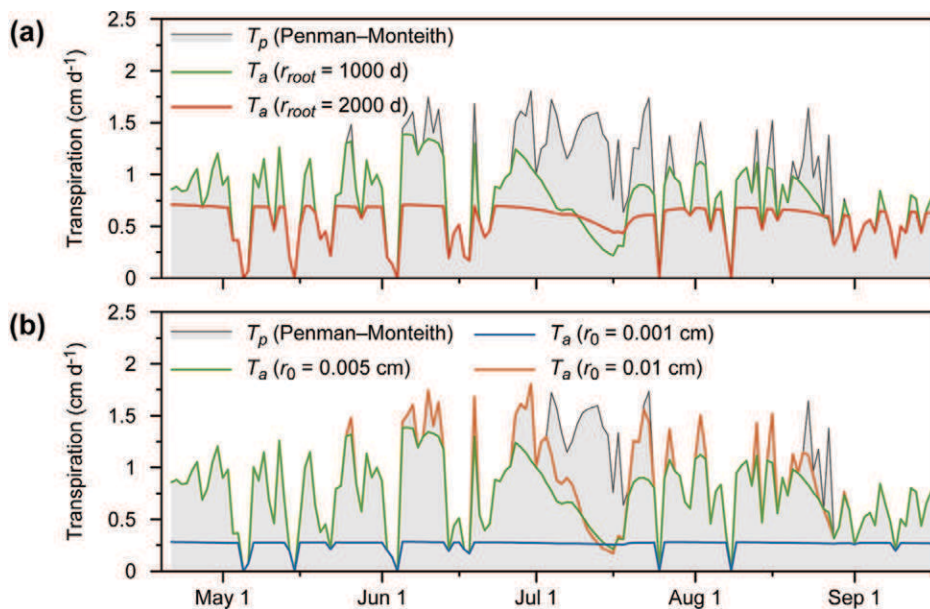


Fig. 8. Actual transpiration predicted by the water-potential-gradient WPG model, based on different values of root water uptake (RWU) parameters, as compared with potential transpiration estimated from micrometeorological data: (a) the impact of different root radial hydraulic resistance r_{root} , with r_0 fixed at 0.005 cm; (b) the impact of changing specific active root surface via variation of r_0 in Eq. [18], with r_{root} fixed at 1000 d. The transpiration rates T_a and T_p are plotted as daily maxima.

Table 2. Statistical comparison between observed and simulated soil water state variables (pressure and water content) at two depths below the soil surface (15 and 30 cm). The last column shows the predicted reduction of the potential transpiration over the vegetation season 2010 (calculated as cumulative value of $(T_p - T_a)/T_p$, where the cumulative value of T_p was estimated to be 352 mm).

Root water uptake (RWU) model	RWU parameters		Soil water pressure				Soil water content				Reduction of T_p
	r_0	r_{root}	$R^2_{\uparrow}$		d		R^2		d		
			15 cm	30 cm	15 cm	30 cm	15 cm	30 cm	15 cm	30 cm	
	cm	d									%
Water-potential-gradient (WPG)	0.005	1000	0.807	0.537	0.903	0.660	0.620	0.733	0.875	0.789	13
	0.001	1000	0.501	0.653	0.417	0.519	0.404	0.542	0.549	0.627	58
	0.010	1000	0.780	0.506	0.737	0.523	0.664	0.740	0.871	0.753	9
	0.005	2000	0.780	0.547	0.797	0.830	0.577	0.740	0.766	0.859	25
Feddes	–	–	0.653	0.297	0.598	0.066	0.583	0.665	0.856	0.716	6

† R^2 is the coefficient of determination, d is the Willmott index of agreement (Willmott, 1981).

local RWU decrease relates to locally overdried soil predicted by this simulation (Fig. 4). The RWU development after the rainfall event at the end of the dry spell in July reflects the soil water replenishment (transition of the upper boundary of reduced RWU region). At the same time, the process of overdrying continues at deeper layers and RWU reduction at lower part of the third layer propagates until the end of simulated period.

The impact of the simulated RWU on the soil water distribution predictions is demonstrated in Fig. 4 and 5. An improved agreement between the simulated and measured soil water pressure values for the WPG model compared to the Feddes model, especially during the dry period in the first half of July is clear. In the Feddes simulation, the third soil layer remains unrealistically dry until the end of the simulated period (Fig. 5b); this is even more pronounced below 30-cm depth (obvious from RWU reduction pattern in Fig. 3b). The observed and simulated soil water state variables (pressure and water content) are statistically compared in Table 2.

Comparison with Sap Flow Measurements

The predicted actual transpiration of spruce trees is compared with the measured HFD sap flow data in Fig. 7a. The two variables are compared in relative terms only as the sap flow data characterize responses of an individual spruce specimen. Application of an upscaling procedure would be necessary to compare these numbers with the transpiration per unit ground area in absolute terms. Since the upscaling of transpiration data from single trees to forest stand would introduce additional uncertainties (Cermak and Nadezhkina, 1998), it was not employed in this study. The differences between the daily variations as observed and simulated are obvious; the simulated transpiration is concentrated into daytime hours whereas in reality non-zero nightly transpiration and/or nightly plant xylem water storage replenishment is observed. Moreover, negative nocturnal sap flow episodes indicating the xylem storage discharge could not be captured in simulation.

Nightly Hydraulic Redistribution

Experimental evidence for night transpiration in woody plants was found, for example, by Dawson et al. (2007). A relationship between the magnitude of nighttime hydraulic redistribution and the intensity of night transpiration was studied e.g., by Howard et al. (2009). In our study, significant nighttime sap flow rates in spruce trees were detected by HFD measurements (with mean estimated fraction of nighttime to daytime rates 0.17 over the vegetation season). These nightly rates can be partly attributed to changes in plant water storage overnight (not considered in our model) and partly to night transpiration (also neglected in our model). The WPG model predicted regular nightly redistribution of soil water (Fig. 3d). The predicted rates are somewhat exaggerated due to the assumed zero transpiration rates overnight. In reality the redistribution would be less pronounced because of non-zero nightly transpiration and/or nightly plant xylem water storage replenishment. On the other hand, occasional negative nocturnal sap flow episodes indicating xylem storage discharge (Fig. 7a), might lead to stronger hydraulic redistribution.

Daytime Hydraulic Redistribution

The WPG model also predicted a number of episodes with negative RWU rates during the daylight hours (Fig. 3d), indicating daytime hydraulic redistribution of soil water mediated by roots. These episodes occurred during rainfall events, when negligible atmospheric vapor pressure deficit and wet canopy effectively stopped transpiration (e.g., on July 24, in Fig. 7). This led to hydraulic lift or descent of soil water depending on the actual moisture status of the affected soil layers.

Simulated Changes of Root Xylem and Root Surface Potentials

The predicted changes of root xylem and root surface water potentials over the selected 12-d period are shown in Fig. 7. The period is situated at the end of a relatively dry part of the season, with heavy rainfall at the beginning of the period (rainfall depth 31

mm) followed by four sunny days and another rain on day seven of the period (24 July, rainfall depth 11 mm). In Fig. 7b, simulated water potentials at 30-cm depth are presented. Water potential difference between the root xylem, H_{root} , and the soil, H_{soil} , can be seen as a driving force of the water exchange between soil and root. The H_{root} values higher than H_{soil} indicate simulated redistribution of water toward the soil at this depth (during zero transpiration conditions). For nonzero transpiration, the instances when H_{root} reaches the limiting value $H_{\text{lim}} = -150$ m indicate reduction of the transpiration rate. It can be seen that these situations occur even after the soil water potential increases above -4 m (indicating moderately dry conditions (cf. Fig. 4)). This illustrates that in the WPG model the transpiration rate is determined based on the water status throughout the whole root zone. Additionally, simulated water potential at the root surface, H_{surf} , is presented (Fig. 7b). Its relative position within the $(H_{\text{soil}}, H_{\text{root}})$ interval indicates the relative proportion of actual resistance to water exchange exhibited by either root or soil (note logarithmic scale of the water potential axis). It can be seen that this proportion changes with soil water content (expressed by H_{soil}); as soil water content rises, the soil resistance decreases (root resistance does not change in time). It should be noted that the predicted root surface water potential H_{surf} exhibits a significant spatiotemporal variability (Fig. 7c).

WPG Model Parameters

The WPG model is defined by four parameters [$R(z)$, r_0 , r_{root} , and H_{lim}]. All these parameters are rather difficult to evaluate; still, their approximate values can be found in literature. However, in our case, adopting published values of r_{root} (on the order of 10,000 d) led to an unrealistic reduction of the transpiration rate (for the site given). To obtain a simulation predicting a reasonable degree of transpiration reduction, we decided to decrease the r_{root} value to 1000 d. The sensitivity of the simulated actual transpiration to the magnitude of r_{root} is demonstrated in Fig. 8a. Inspection of Eq. [14] shows that similar adjustment of transpiration reduction could be achieved by manipulating the active root surface σ , which is a function of r_0 . The impact of changing r_0 is demonstrated in Fig. 8b and Table 2. It can be seen that, unlike in de Jong van Lier et al. (2008), our modeling results are quite sensitive to the change of active root radius r_0 . Obviously, it should be possible to find another combination of r_{root} and r_0 values that would produce similar simulation results as those presented; however, increasing r_{root} would lead to further diminishing of the soil hydraulic conductivity control of root water uptake. The relative magnitude of the resistances r_{root} and r_{soil} is illustrated in Fig. 9 (based on $r_{\text{root}} = 1000$ d and $r_0 = 0.005$ cm). Most of the time, r_{root} dominated over r_{soil} , except during the dry period where the relationship reversed. If r_{root} was higher, r_{soil} would become negligible during most of the simulated period. Although we deem our simulation results reasonable, the WPG model parameter values should be considered as estimates open to discussion.

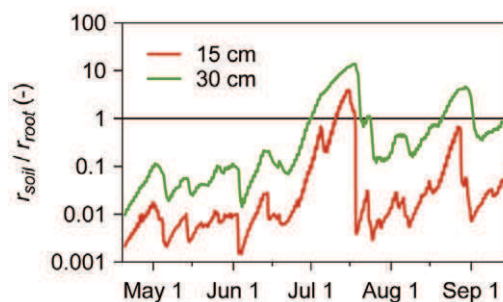


Fig. 9. The $r_{\text{soil}}/r_{\text{root}}$ ratio obtained with the water-potential-gradient (WPG) model ($r_{\text{root}} = 1000$ d, $r_0 = 0.005$ cm).

WPG Model Adequacy

The WPG algorithm in its present form is a highly simplified mechanistic representation of real root water uptake dynamics. It does not take into account the moderating effects of plant xylem water storage and longitudinal xylem resistance; it ignores the intricacies of stomatal control of the transpiration stream, and utilizes a single-valued threshold of the root xylem potential to trigger the reduction of potential transpiration.

On the other hand, there are several benefits of this approach, such as: simple and well defined parameterization, explicit handling of distributed flow resistances between root xylem and bulk soil, natural inclusion of gravitational potential (and possibly also osmotic potential), and easy implementation into existing water potential based numerical models of soil water movement.

Several model assumptions may be easily relaxed, e.g.: root radial hydraulic resistance, r_{root} , does not need to be constant with depth (or with time), r_{root} may be further parameterized to reflect the possible dependency on local RWU rate, soil water status, solute concentration, or temperature. The reduction procedure based on single-valued threshold root xylem potential H_{lim} can be replaced by a more realistic plant response function. However, all these model extensions require additional information unavailable in our study.

Summary and Conclusions

A simple macroscopic vertically distributed plant root water uptake model based on traditional water-potential-gradient formulation, in which the uptake rates are directly proportional to the potential gradient and indirectly proportional to the local soil and root resistances to water flow, was tested in a case study. The model was capable of simulating both compensatory root water uptake, in situations when reduced uptake from dry layers was compensated for by increased uptake from wetter layers, and the root-mediated hydraulic redistribution of soil water, contributing to a more natural soil moisture distribution throughout the root zone.

Comparison of the model results with the sap flow observations revealed some limitations related to the quasi-steady state assumption for the plant xylem and zero transpiration rates prescribed during nights and precipitation. This stated, the model seems to simulate adequately both the regular nightly hydraulic redistribution, due to reduced night transpiration, and the episodic daytime hydraulic redistribution during wet canopy events.

The model results were compared to simulations produced using the semi-empirical RWU model of Feddes. Based on both an improved agreement between the observed and simulated soil water pressure responses to daily variations of transpiration, and a more realistic seasonal distribution of the transpiration rate reduction, we concluded that the physically based root water uptake model with possibly negative RWU rates enabled substantially better approximation of the soil water extraction by spruce trees under moderate water scarcity.

Acknowledgments

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